

1 Solution — GIAM 3.3

Problem 3

1.1 Problem

Prove (by contradiction) that there is no largest integer.

1.2 Formalising the Claim

“ N is the largest integer” means $N \in \mathbb{Z}$ and every integer is at most N . So the theorem is

$$\neg \left(\exists N \in \mathbb{Z}, \forall x \in \mathbb{Z}, x \leq N \right),$$

whose negation — the thing we assume for contradiction — is

$$\exists N \in \mathbb{Z}, \forall x \in \mathbb{Z}, x \leq N.$$

1.3 Answer

No such N exists: $N + 1$ is an integer exceeding N .

1.4 Proof

Theorem. *There is no largest integer.*

Proof. *Suppose, to the contrary, that there is a largest integer. Call it N . Then $N \in \mathbb{Z}$, and*

$$\forall x \in \mathbb{Z}, \quad x \leq N.$$

Since N and 1 are integers and $\mathbb{Z}$ is closed under addition, $N + 1$ is an integer. So $N + 1$ is a legitimate choice for x in the statement above, giving

$$N + 1 \leq N.$$

Subtracting N from both sides yields $1 \leq 0$, which is false.

The supposition that a largest integer exists leads to a contradiction, so there is no largest integer. ■

1.5 The Picture

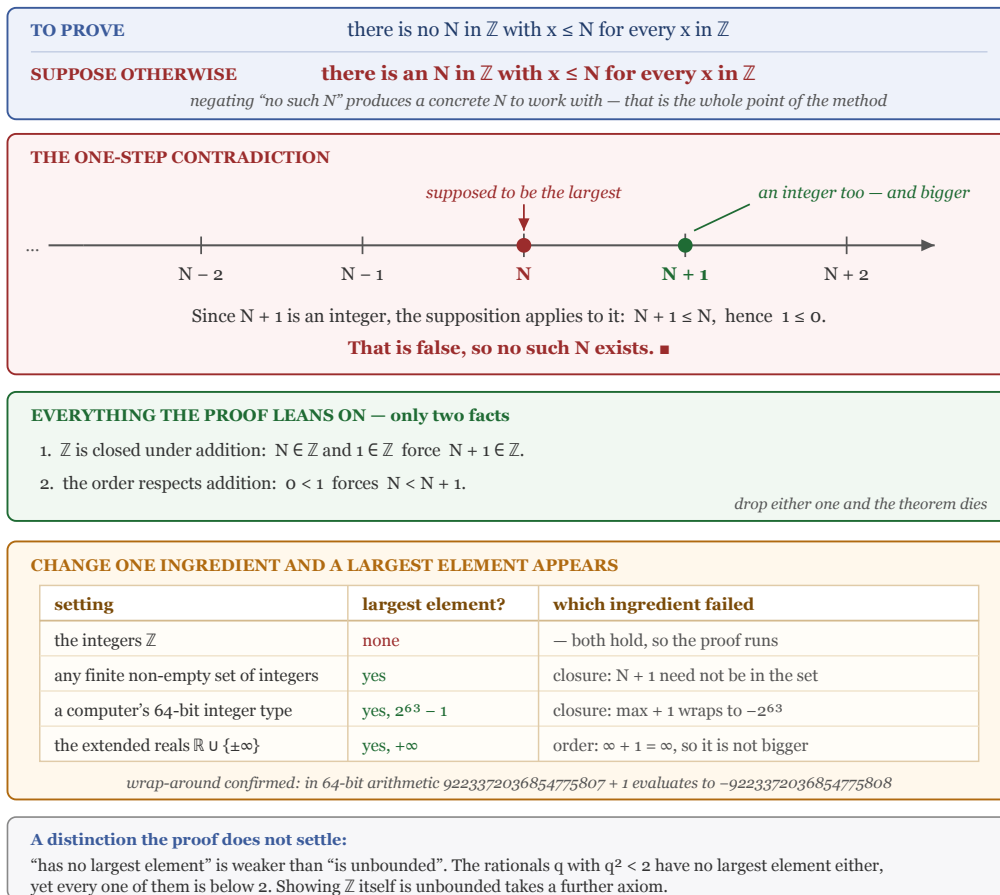


Figure 1.1: A diagram in five bands. The first states what is to be proved — there is no N in the integers with x at most N for every integer x — and then the supposition otherwise, that such an N exists, noting that negating ‘no such N ’ produces a concrete N to work with, which is the whole point of the method. The second band shows a number line marked N minus 2, N minus 1, N , N plus 1, N plus 2, with N in red flagged ‘supposed to be the largest’ and

N plus 1 in green flagged ‘an integer too — and bigger’. Below it: since N plus 1 is an integer, the supposition applies to it, giving N plus 1 at most N , hence 1 at most 0; that is false, so no such N exists. The third band lists everything the proof leans on, only two facts: first, the integers are closed under addition, so N and 1 being integers forces N plus 1 to be an integer; second, the order respects addition, so 0 less than 1 forces N less than N plus 1. Drop either one and the theorem dies. The fourth band shows what happens when one ingredient changes: the integers have no largest element and both facts hold so the proof runs; any finite non-empty set of integers has a largest, because closure fails since N plus 1 need not be in the set; a computer’s 64-bit integer type has a largest, 2 to the 63 minus 1, because closure fails as max plus one wraps to minus 2 to the 63; and the extended reals with plus and minus infinity have a largest, plus infinity, because the order fails as infinity plus one equals infinity and so is not bigger. Wrap-around confirmed: in 64-bit arithmetic 9223372036854775807 plus 1 evaluates to minus 9223372036854775808. The fifth band notes a distinction the proof does not settle: having no largest element is weaker than being unbounded, since the rationals whose square is less than 2 have no largest element either yet all lie below 2, and showing the integers are unbounded takes a further axiom.

1.6 Remark — This “Proof by Contradiction” Is Barely One

The problem asks for contradiction, and the proof above obliges. But look at what actually happens: from the supposed N we build $N + 1$ and observe it is bigger. No feature of the contradiction framework is doing any work — the argument constructs an explicit witness.

Stripping the wrapper off gives a *direct* proof of the same thing:

Proof (direct). Let N be any integer. Then $N + 1$ is an integer with $N + 1 > N$, so N is not the largest. Since N was arbitrary, no integer is the largest. ■

The two are the same three lines. What changed is the logical form: “there is no largest” is a negated existential, and

$$\neg \exists N \forall x (x \leq N) \cong \forall N \exists x (x > N),$$

which is a universally quantified *existence* claim — exactly the sort of thing a direct proof handles by exhibiting the witness $x = N + 1$.

This is a useful thing to notice generally. Many proofs presented as “by contradiction” are direct proofs of the contrapositive or of an equivalent positive statement, dressed up. A genuine proof by contradiction is one where the supposition is used *twice* — once to build something and once to collide with it — as in Euclid’s infinitude of primes, where $N = 1 + \prod p_k$ is constructed *from* the assumed finite list.

1.7 Remark — Every Ingredient the Proof Uses

The argument is short enough that its dependencies can be listed exhaustively:

- **$\mathbb{Z}$ is closed under addition.** This is what makes $N + 1$ an integer, and hence a legal value to substitute for x . Without it, $N + 1$ might live outside the set and the supposition would say nothing about it.
- **The order is compatible with addition,** together with $0 < 1$. This is what makes $N + 1$ strictly bigger than N .

That is all. Notice in particular that *nothing about the integers specifically* is used — the same proof shows $\mathbb{Q}$, $\mathbb{R}$, and the even integers have no largest element, and indeed:

Any ordered set closed under $y \mapsto y + 1$ (or under any operation that strictly increases its input) has no largest element.

1.8 Remark — Where a Largest Element Does Appear

The clean way to appreciate a proof is to see what breaks it. Each of the two ingredients can fail, and each failure produces a set with a genuine maximum:

setting	largest?	what failed
$\mathbb{Z}$	none	— both hold
finite $S \subset \mathbb{Z}$	yes	closure: $N+1$ may leave S
64-bit machine integers	$2^{63} - 1$	closure: $\text{max} + 1$ wraps around
$\mathbb{R} \cup \{\pm\infty\}$	$+\infty$	order: $\infty + 1 = \infty$

The machine-integer row is the most concrete: in 64-bit arithmetic, $9223372036854775807 + 1$ evaluates to -9223372036854775808 , which is confirmed by running it. There *is* a largest `int64`, precisely because the set is not closed under adding one. Arbitrary-precision integers behave like $\mathbb{Z}$, and the $N + 1$ witness works at any size.

The extended-reals row fails differently and is worth pausing on: $\mathbb{R} \cup \{\pm\infty\}$ *is* closed under adding 1, but $\infty + 1 = \infty$, so the successor is not strictly larger. Both ingredients really are needed.

1.9 Remark — No Maximum Versus Unbounded

It is tempting to read this theorem as “the integers go on forever”, but strictly it says something weaker. Compare:

$$\begin{array}{l|l} \text{no largest element} & \text{no } N \in S \text{ with } x \leq N \text{ for all } x \in S \\ \text{unbounded above} & \text{no } B \text{ at all with } x \leq B \text{ for all } x \in S \end{array}$$

The second is stronger, because the bound B need not belong to S . A set can miss its own ceiling:

$$S = \{q \in \mathbb{Q} : q^2 < 2\}$$

has no largest element — for any $q \in S$ there is a larger rational still in S — yet every element is below 2. Being maximum-free is not the same as being boundless.

For $\mathbb{Z}$ inside $\mathbb{R}$ the stronger statement *is* true, but it needs more than the $N + 1$ trick: it is the **Archimedean property**, which follows from the completeness of $\mathbb{R}$ (a bounded $\mathbb{Z}$ would have a least upper bound s , and $s - 1$ would fail to be one). In a non-Archimedean ordered field — the hyperreals, say — some element exceeds every integer, so there $\mathbb{Z}$ has no largest element yet *is* bounded above. The theorem survives; the intuitive reading of it does not.